

Question	Answer	Marks
3(a)(i)	work (done)/time (taken)	B1
3(a)(ii)	energy of a mass due to its position in a gravitational field	B1
3(b)(i)	$P = Fv$	C1
	$= 2.0 \times 10^3 \times 45$ $= 9.0 \times 10^4 \text{ W}$	A1
3(b)(ii)	1. $W = (2.0 \times 10^3) \times (45 \times 3.0 \times 60)$ or $W = 9.0 \times 10^4 \times 3.0 \times 60$	C1
	$W = 1.6 \times 10^7 \text{ J}$	A1
	2. $(\Delta)E_P = mg(\Delta)h$	C1
	$= 1200 \times 9.81 \times 3.3 \times 3.0 \times 60$ $= 7.0 \times 10^6 \text{ J}$	A1
	3. $W = 1.6 \times 10^7 - 7.0 \times 10^6$ $= 9.0 \times 10^6 \text{ J}$	A1
3(b)(iii)	force $= (9.0 \times 10^6) / (45 \times 3.0 \times 60)$ $= 1.1 \times 10^3 \text{ N}$	A1
3(b)(iv)	constant velocity so no resultant force	B1
	no resultant force so in equilibrium	B1

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4(c)	when (two or more) waves meet (at a point)	D1